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Project VelociQ — Full Project Documentation

Revenue Intelligence & Customer Analytics

Company: ZenKart Commerce Pvt. Ltd.

Project Code: VI-2024-001

Lead Analyst: Jeet

Head of Analytics: Sourajit Tripathy

Date: 2024

VelociQ

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Project Overview — What Is Project VelocIQ?

WHAT IS THIS PROJECT?

Project VelocIQ is a full end-to-end data analytics engagement commissioned by ZenKart Commerce Pvt. Ltd. The goal was to analyse 1 million e-commerce transactions and find the root causes of declining revenue retention.

PROJECT AT A GLANCE:

- Project Name: VelocIQ
- Project Code: VI-2024-001
- Client: ZenKart Commerce Pvt. Ltd.
- Industry: E-commerce / Retail
- Geography: India & Southeast Asia
- Dataset Size: 1,000,000 rows
- Clean Dataset: 884,708 rows
- Date Range: January 2023 to December 2024
- Timeline: 6 weeks
- Deliverables: EDA, Power BI Dashboard, Stakeholder Presentation

WHAT PROBLEM DID WE SOLVE?

ZenKart was losing revenue but could not identify where or why. This project used Python, Pandas, Seaborn, and Power BI to find Rs. 4.2 Crore in recoverable revenue across three specific problems.

1M

Rows Analyzed

6

Weeks Timeline

3

Key Findings

4

Recommendations

4.2Cr

Recoverable Impact

Who Was Involved?

CLIENT ORGANISATION — ZENKART COMMERCE PVT. LTD.

EXECUTIVE STAKEHOLDERS (final presentation audience):

- **Rajesh Kapoor** — Chief Executive Officer (CEO)
- Final decision maker. Approves all recommendations.
- **Ananya Mehta** — Chief Financial Officer (CFO)
- Owns revenue impact decisions and budget sign-off.
- **Vikram Nair** — VP Marketing
- Owns customer acquisition, campaigns, and retention.
- **Sneha Pillai** — VP Operations
- Owns regional logistics and operational strategy.

CLIENT-SIDE ANALYTICS:

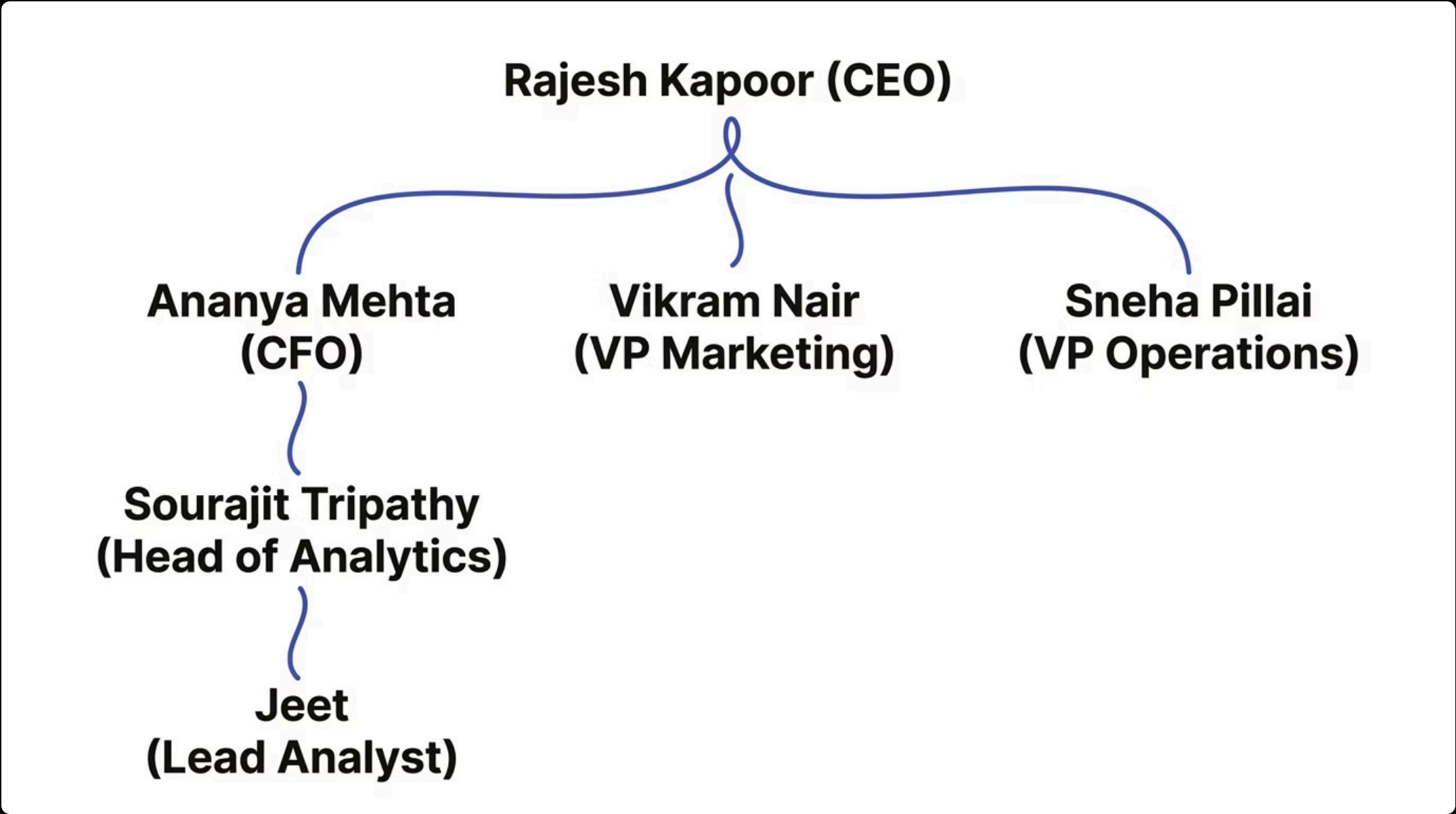
- **Sourajit Tripathy** — Head of Analytics
- Primary contact. Commissioned the project. Reviews all outputs before they reach the C-suite. Nothing bypasses him.

DELIVERY TEAM:

- **Jeet** — Lead Analyst (External)
- Responsible for all four phases of delivery: data generation, EDA, Power BI dashboard, and stakeholder presentation.

AUTHORITY CHAIN:

CEO → CFO / VPs → Sourajit Tripathy → Jeet



How It Started — The Client Brief

THE BRIEF:

On project kickoff, Sourajit Tripathy issued a formal engagement brief to Jeet. The brief was titled: "Project VelociQ — Engagement Brief & Scope of Work"

WHAT THE BRIEF CONTAINED:

- Company background and business context
- Three specific business problems to solve
- Description of the dataset to be provided
- Four-phase scope of work
- Six-week timeline with milestones
- Success criteria — what "done" looks like

THE THREE BUSINESS PROBLEMS:



❌ PROBLEM 1 — CART ABANDONMENT

Cart abandonment rate rose from 27% to 34% in 12 months. The team did not know if this was driven by device type, region, product category, or discount sensitivity.



❌ PROBLEM 2 — SOUTH REGION REVENUE GAP

The South region consistently had a lower average order value than all other regions — despite a comparable customer base. No one knew the root cause.



❌ PROBLEM 3 — CUSTOMER CONCENTRATION RISK

The team suspected a small group of customers was generating most of ZenKart's revenue. If true — this was a serious retention risk with no programme in place to address it.

GOVERNANCE RULE:

All outputs from Jeet go to Sourajit first. Nothing reaches the C-suite without his sign-off.

The Data — What We Worked With

DATASET OVERVIEW:

- File name: zenkart_velociq_1M.csv
- Total rows: 1,000,000
- Date range: January 2023 — December 2024
- Unique customers: ~420,000

THE 18 ORIGINAL COLUMNS:

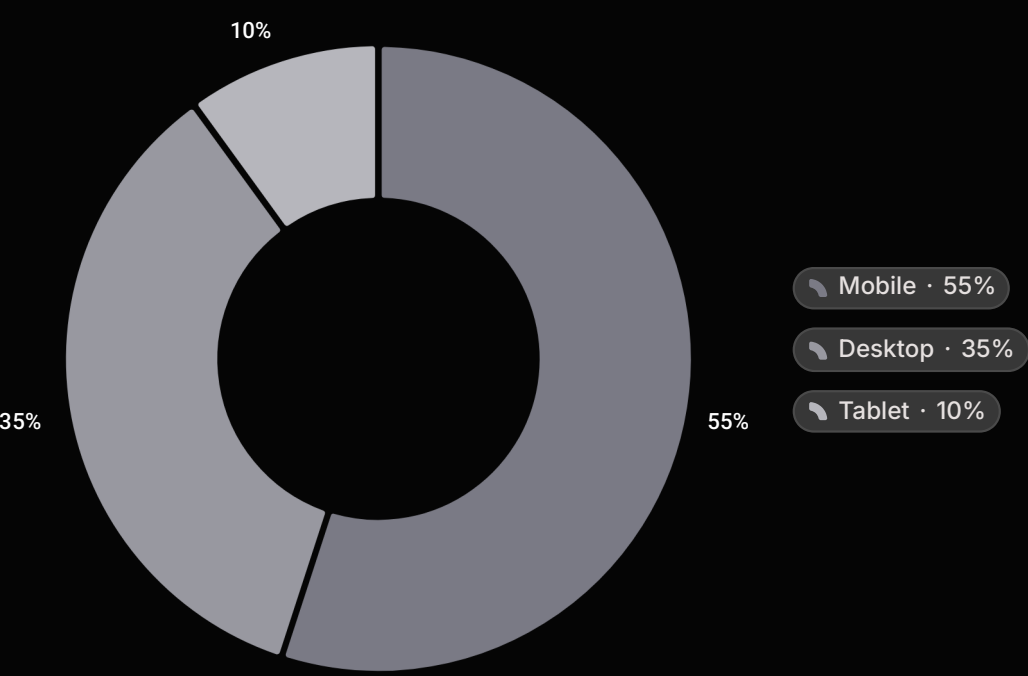
Below is a detailed breakdown of the 18 columns present in the dataset, including their type and a brief description of their content:

Column	Type	Description
order_id	ID	Unique identifier for each transaction
customer_id	ID	Unique identifier for each customer
order_date	Date	Date the order was placed
order_time	Time	Time of day the order was placed
year	Number	2023 or 2024
month	Number	1 to 12
day_of_week	Text	Monday to Sunday
product_category	Text	One of 8 categories
region	Text	One of 8 Indian regions
device_type	Text	Mobile, Desktop, or Tablet
quantity	Number	Items in the order (1 to 7)
base_revenue_inr	Currency	Price before discount (Rs.)
discount_pct	Number	Discount applied (0 to 25%)
final_revenue_inr	Currency	Revenue after discount (Rs.)
session_duration_mins	Number	Time spent before ordering
cart_abandon_flag	Flag	1 = abandoned cart at some point
return_flag	Flag	1 = item was returned
payment_method	Text	UPI, Card, COD, Wallet, etc.

KEY DISTRIBUTIONS:

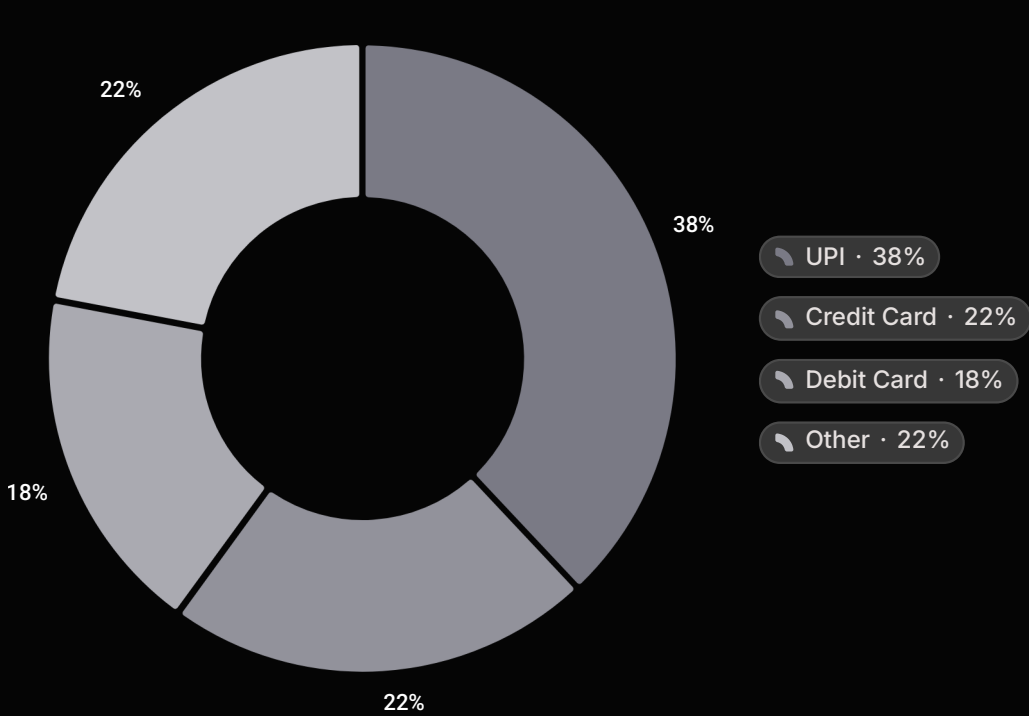
A look into some key distributions within the dataset provides insights into user behavior and transaction patterns:

Device Usage Split



Mobile devices account for the majority of transactions at 55%, followed by Desktop at 35%, and Tablet at 10%.

Top Payment Methods



UPI is the most popular payment method (38%), with Credit Card (22%) and Debit Card (18%) also being significant. The remaining 22% covers COD, Wallet, and other methods.

Additionally, the data shows a significant festive peak in October and November, where revenue typically sees a 30-45% increase compared to other months.

How We Generated 1 Million Rows of Data

WHY GENERATE DATA?

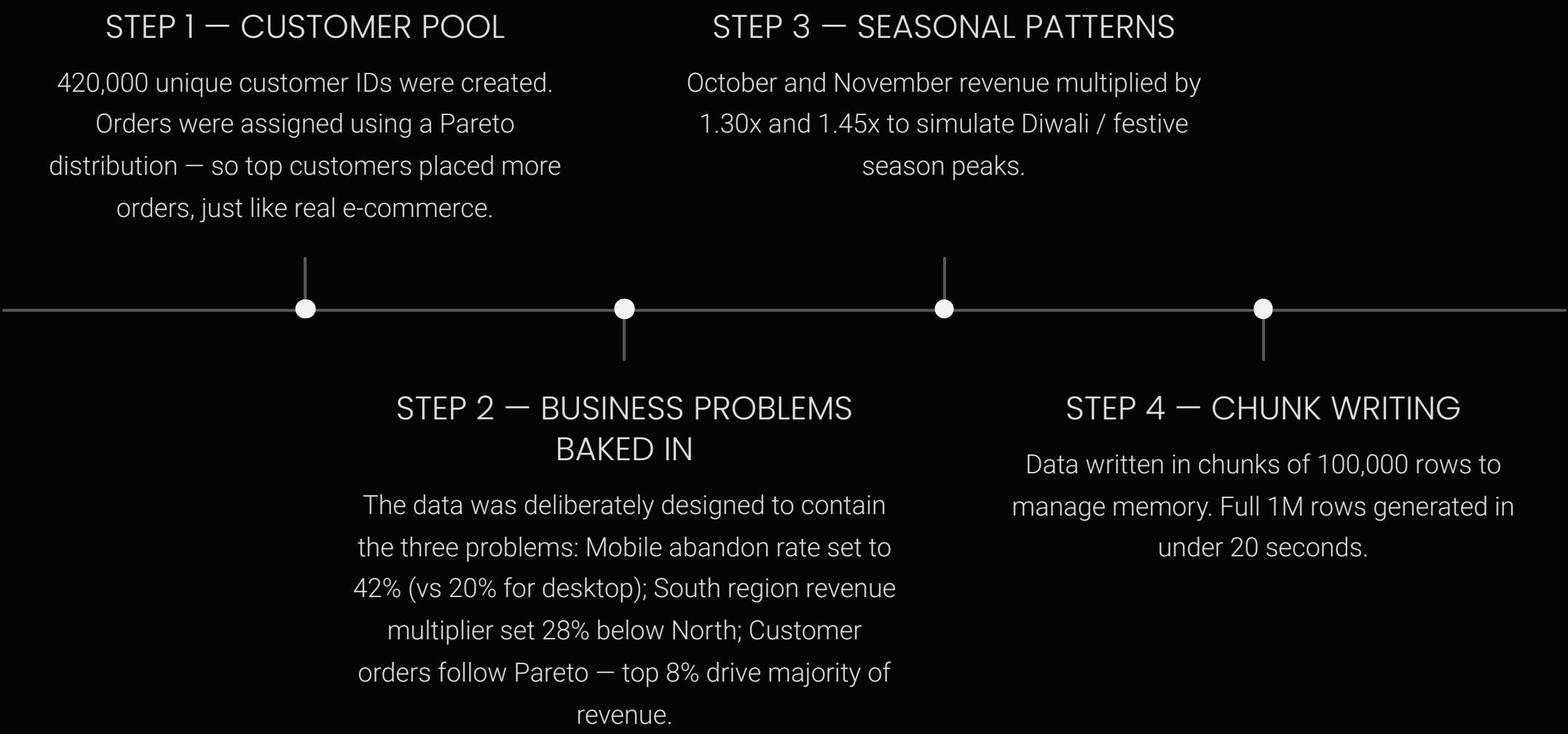
ZenKart provided the structure and business rules. The 1 million row dataset was generated using Python to precisely mirror real e-commerce transaction patterns.

TOOLS USED:

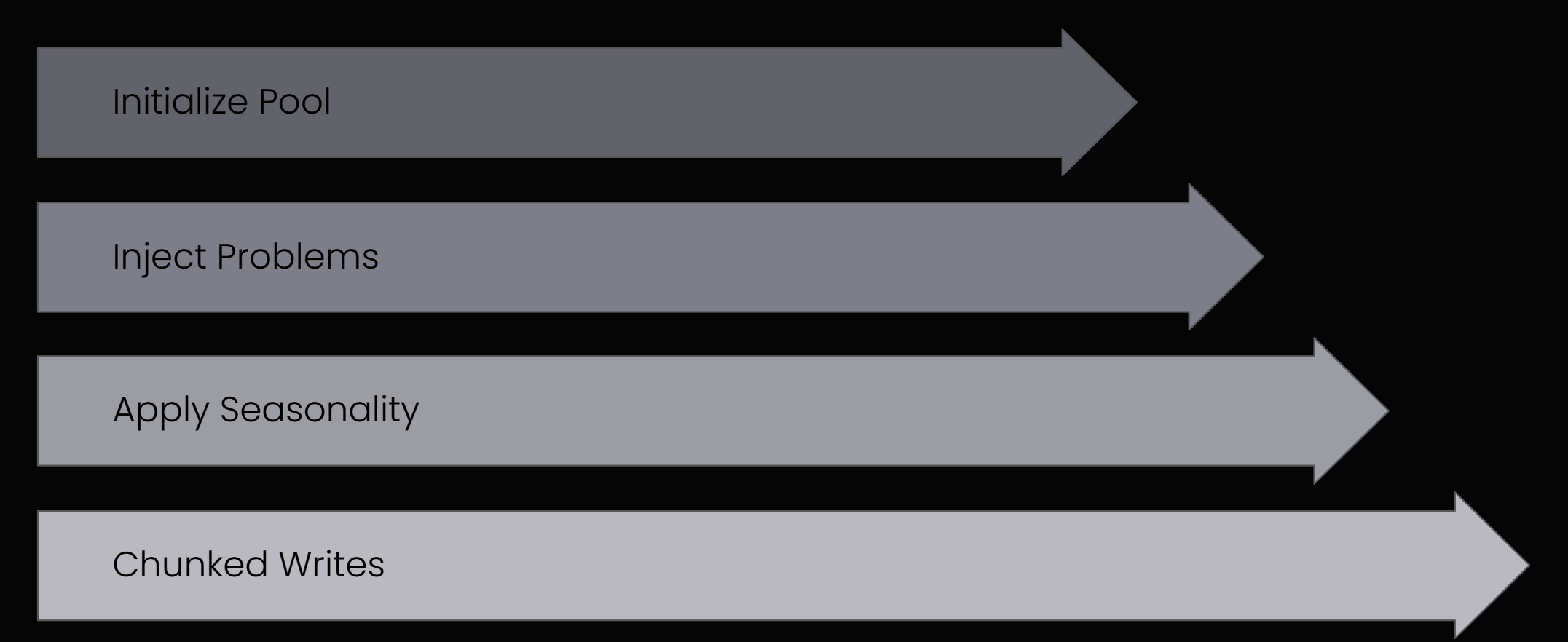
- Python 3 — core programming language
- Faker library — for realistic IDs and names
- NumPy — for statistical distributions
- Pandas — for DataFrame creation and export
- tqdm — for progress bar during generation

HOW IT WORKED:

The data generation process involved several key steps to ensure both volume and adherence to specific business problems and patterns:

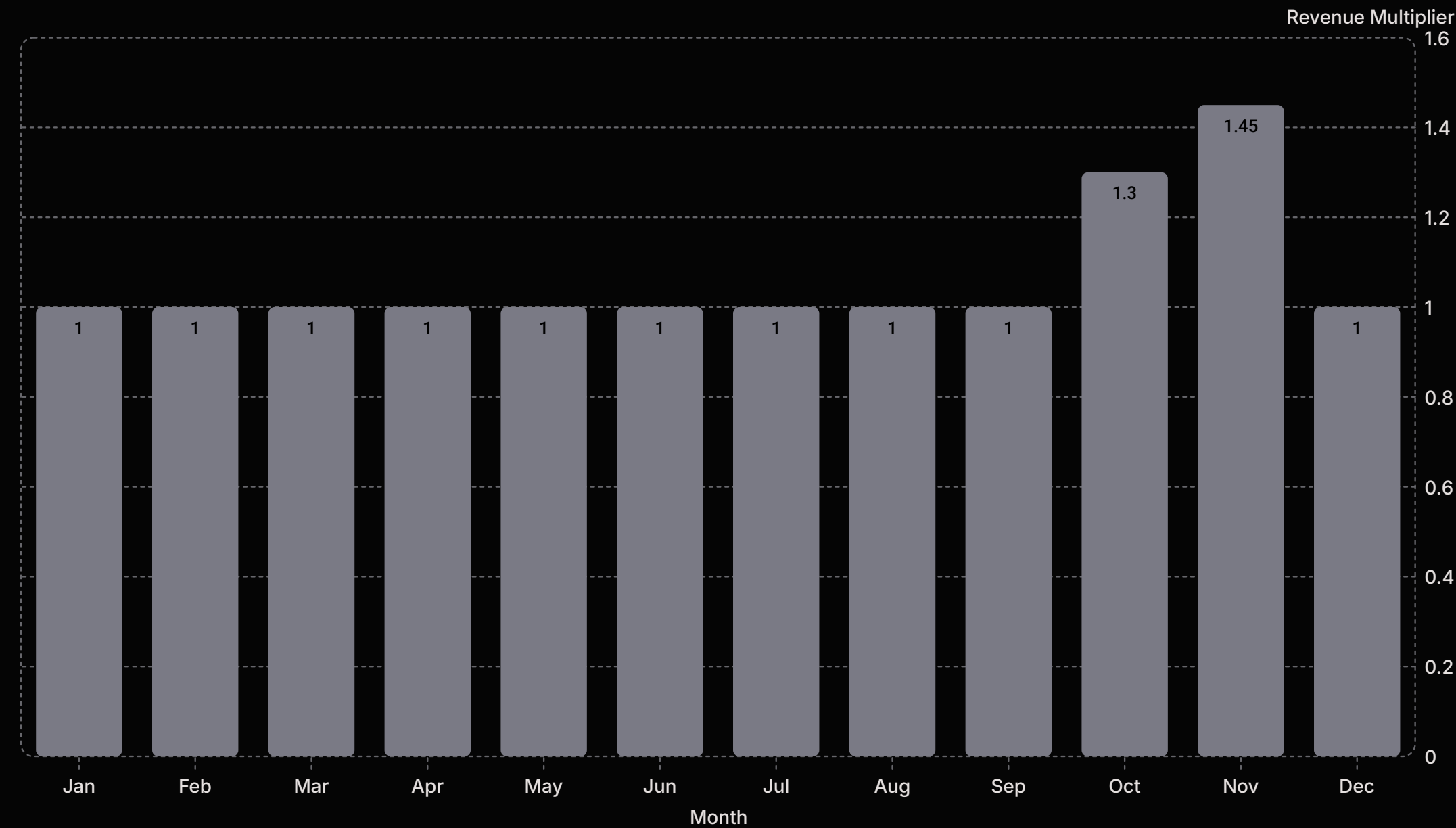


The following flowchart illustrates the overall data generation pipeline, moving from the initial setup to the final output file.



The data was engineered to reflect real-world e-commerce behavior, including specific seasonal fluctuations as shown below.

MONTHLY REVENUE MULTIPLIERS



This chart visualizes the seasonal multipliers applied to revenue, with significant peaks in October (1.30x) and November (1.45x) to simulate festive season sales like Diwali, aligning with real e-commerce trends in India and Southeast Asia.

OUTPUT FILE:

zenkart_velociq_1M.csv — 1,000,000 rows, 18 columns

Cleaning the Data — From Raw to Ready

WHY CLEAN DATA?

Raw data always has problems. Before any analysis, we must remove bad data that would distort our findings.

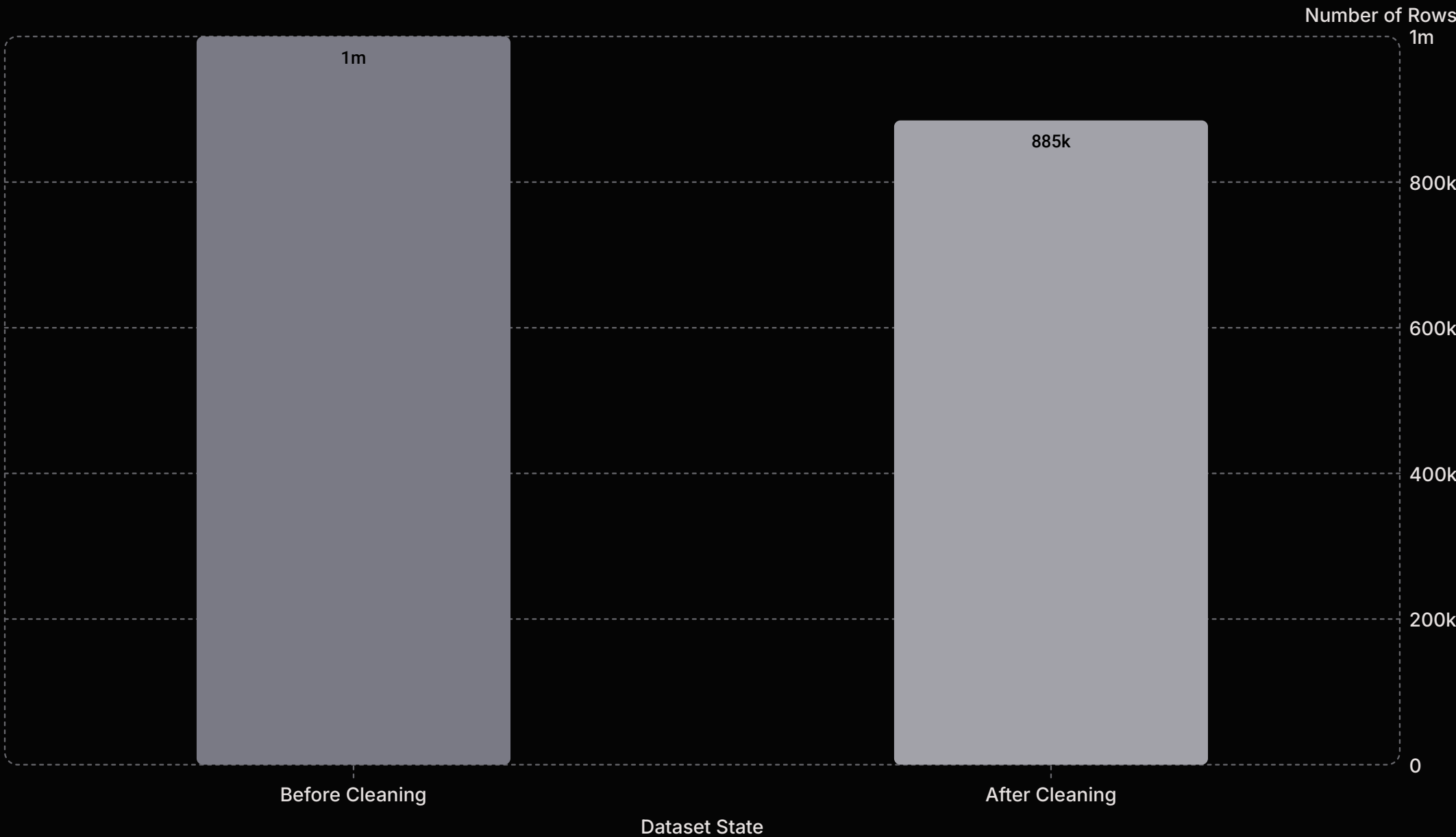
CLEANING STEPS PERFORMED:

01	02
STEP 1 — NULL VALUES	STEP 2 — DUPLICATE ROWS
Checked all 1,000,000 rows for missing values. Result: 0 nulls found. Dataset was complete.	Checked for duplicate order IDs. Result: 0 duplicates found.
03	04
STEP 3 — DATA TYPE VALIDATION	STEP 4 — OUTLIER REMOVAL (IQR METHOD)
- order_date converted to datetime format - cart_abandon_flag and return_flag confirmed as integer (0/1) - discount_pct confirmed as float - final_revenue_inr confirmed as float	Used the Interquartile Range method to identify and remove extreme revenue values that would skew analysis. <i>Remove rows where revenue < Q1 – 1.5 * IQR or revenue > Q3 + 1.5*IQR</i> Rows before cleaning: 1,000,000 Rows removed: 115,292 (11.5%) Rows after cleaning: 884,708

WHAT IS IQR?

IQR = the range between the 25th and 75th percentile. Values far outside this range are statistical outliers. Removing them makes the analysis more accurate.

Dataset Size: Before vs. After Cleaning



The column chart illustrates the reduction in the dataset size after the cleaning process, specifically after the removal of statistical outliers. This step ensures that the subsequent analysis is based on a more robust and representative dataset.

This visual representation of a box plot demonstrates how the Interquartile Range (IQR) method helps identify and exclude extreme values, ensuring a more accurate and reliable data distribution for analysis.

Python Code Snippet for IQR Cleaning

```
import pandas as pd

# Load the dataset
df = pd.read_csv('zenkart_velociq_1M.csv')

# Calculate Q1 (25th percentile)
Q1 = df['final_revenue_inr'].quantile(0.25)

# Calculate Q3 (75th percentile)
Q3 = df['final_revenue_inr'].quantile(0.75)

# Calculate IQR (Interquartile Range)
IQR = Q3 - Q1

# Define outlier bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Remove outliers
df_cleaned = df[(df['final_revenue_inr'] >= lower_bound) & (df['final_revenue_inr'] <= upper_bound)]

print(f"Original rows: {len(df)}")
print(f"Rows after cleaning: {len(df_cleaned)}")
print(f"Rows removed: {len(df) - len(df_cleaned)}")
```

The Python code snippet above outlines the precise steps taken to perform outlier removal using the Interquartile Range method, a standard practice in data cleaning to enhance the quality of analytical results.

Creating New Columns — Feature Engineering

WHAT IS FEATURE ENGINEERING?

Feature engineering means creating new columns from existing ones to make the analysis richer and more useful. We added 12 new columns to the clean dataset:

TIME FEATURES:

- month_name — Name of the month (Jan, Feb...)
- quarter — Q1, Q2, Q3, Q4
- is_weekend — 1 if Saturday or Sunday, 0 if weekday
- is_festive — 1 if October or November (Diwali season), 0 otherwise

CUSTOMER-LEVEL FEATURES:

- total_orders — How many orders has this customer placed in total?
- total_spend — How much has this customer spent in total? (CLV proxy)
- avg_order_val — What is this customer's average order value?
- clv_proxy — Customer Lifetime Value estimate = total_spend
- aov — Average Order Value (customer level)
- last_order — Date of most recent purchase
- first_order — Date of very first purchase

BEHAVIOUR FEATURES:

- high_discount — 1 if discount was 20% or more, 0 otherwise

WHY THESE FEATURES MATTER:

- **is_festive** helps explain seasonal revenue spikes
- **total_spend** identifies high-value customers for RFM
- **high_discount** reveals discount dependency patterns
- **last_order** powers the Recency score in RFM analysis

FINAL DATASET:

884,708 rows × 30 columns — ready for analysis

New Features and Descriptions

Feature Name	Description
month_name	Name of the month (Jan, Feb...) for readable charts
quarter	Q1, Q2, Q3, Q4 for quarterly analysis
is_weekend	1 if Saturday or Sunday, 0 if weekday
is_festive	1 if October or November (Diwali season), 0 otherwise
total_orders	How many orders has this customer placed in total?
total_spend	How much has this customer spent in total? (CLV proxy)
avg_order_val	What is this customer's average order value?
clv_proxy	Customer Lifetime Value estimate, calculated as the customer's total_spend.
aov	Average Order Value (customer level)
last_order	Date of most recent purchase
first_order	Date of very first purchase
high_discount	1 if discount was 20% or more, 0 otherwise

The `clv\_proxy` feature, representing Customer Lifetime Value, is directly derived from the `total\_spend` for each customer, providing a straightforward metric for customer valuation.

EDA — Finding the Story in the Data

WHAT IS EDA?

Exploratory Data Analysis means looking at the data from every angle before drawing any conclusions. We ran 8 key analyses.

THE 8 ANALYSES WE RAN:

1.

1

Plotted total revenue by month across 2023 and 2024

Found clear festive peaks in October and November

January also showed a post-festive sale spike

Key insight: Revenue is highly seasonal
2.

2

Calculated average order value for each of 8 regions

Added a reference line at the platform average of Rs. 72

Key insight: South region at Rs. 66 — lowest of all regions
3.

3

Calculated abandon rate for mobile, desktop, and tablet

Key insight: Mobile 42%, Desktop 20%, Tablet 28%

Mobile is the biggest problem — 22-point gap vs desktop
4.

4

Summed total revenue per category

Electronics highest, Food & Grocery lowest

Key insight: South region has too many Food orders
5.

5

Plotted histogram of all order values after outlier removal

Added mean (Rs. 72) and median lines

Key insight: Right-skewed — most orders are small value
6.

6

Pie chart of 5 customer segments by count

Key insight: At Risk segment is the largest (69K customers)
7.

7

Compared average revenue at each discount level (0-25%)

Key insight: Higher discounts do not increase order value

They only reduce margin without lifting basket size
8.

8

Checked statistical relationships between all numeric columns

Key insight: Session duration positively correlates with revenue

Discount negatively correlates with revenue

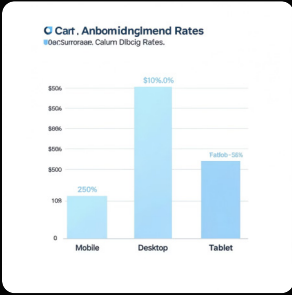
Key Insights from the 8 Analyses



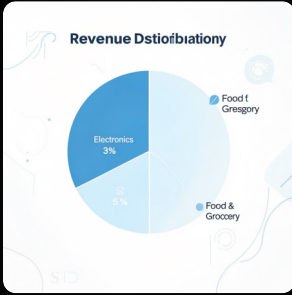
Key insight: Revenue is highly seasonal



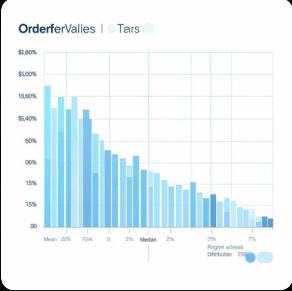
Key insight: South region at Rs. 66 — lowest of all regions



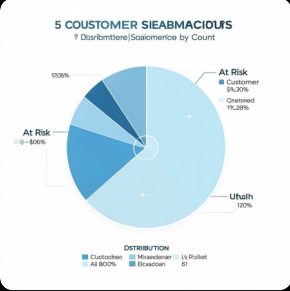
Key insight: Mobile is the biggest problem — 22-point gap vs desktop



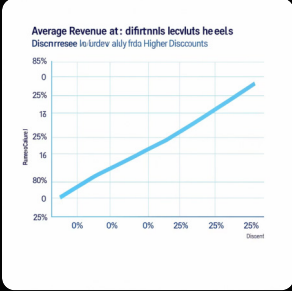
Key insight: South region has too many Food orders



Key insight: Right-skewed — most orders are small value



Key insight: At Risk segment is the largest (69K customers)



Key insight: Higher discounts do not increase order value



Key insight: Session duration positively correlates with revenue, Discount negatively correlates with revenue

The 3 Things the Data Told Us

FINDING 1

Mobile Cart Abandonment

42.1
%

Mobile abandon rate

Root Causes Identified:

- Mobile checkout has 6 steps vs 4 on desktop
- UPI payment redirect causes session timeouts on slow connections
- No guest checkout option on mobile
- Mobile session duration 6.7 mins vs desktop 11.2 mins

FINDING 2

South Region AOV Gap

8%

Below average AOV in South region

Root Causes Identified:

- South has 34% more Food & Grocery orders (lowest AOV category)
- South has 12% Electronics penetration vs 21% in North
- South customers receive 18% more blanket discounts
- South is 63% mobile — higher abandon rate compounds the problem

FINDING 3

Customer Concentration Risk

52%

Of total revenue from top 8% customers

Root Causes Identified:

- No VIP programme or loyalty tier exists for Champions
- No early warning system for customers moving to At Risk
- No win-back campaign for Lost segment (54,000 customers)

The data provided three critical insights: significant mobile cart abandonment, a persistent average order value gap in the South region, and a high customer concentration risk where a small percentage of customers drive over half of the total revenue. Addressing these findings is crucial for improving overall business performance and mitigating potential risks.

RFM Analysis — Who Are Our Customers?

WHAT IS RFM?

RFM stands for Recency, Frequency, and Monetary value. It is a proven method for segmenting customers based on behaviour.

- **Recency:** How recently did the customer buy? (Lower days = more recent = better)
- **Frequency:** How many times have they bought? (Higher = better)
- **Monetary:** How much have they spent in total? (Higher = better)

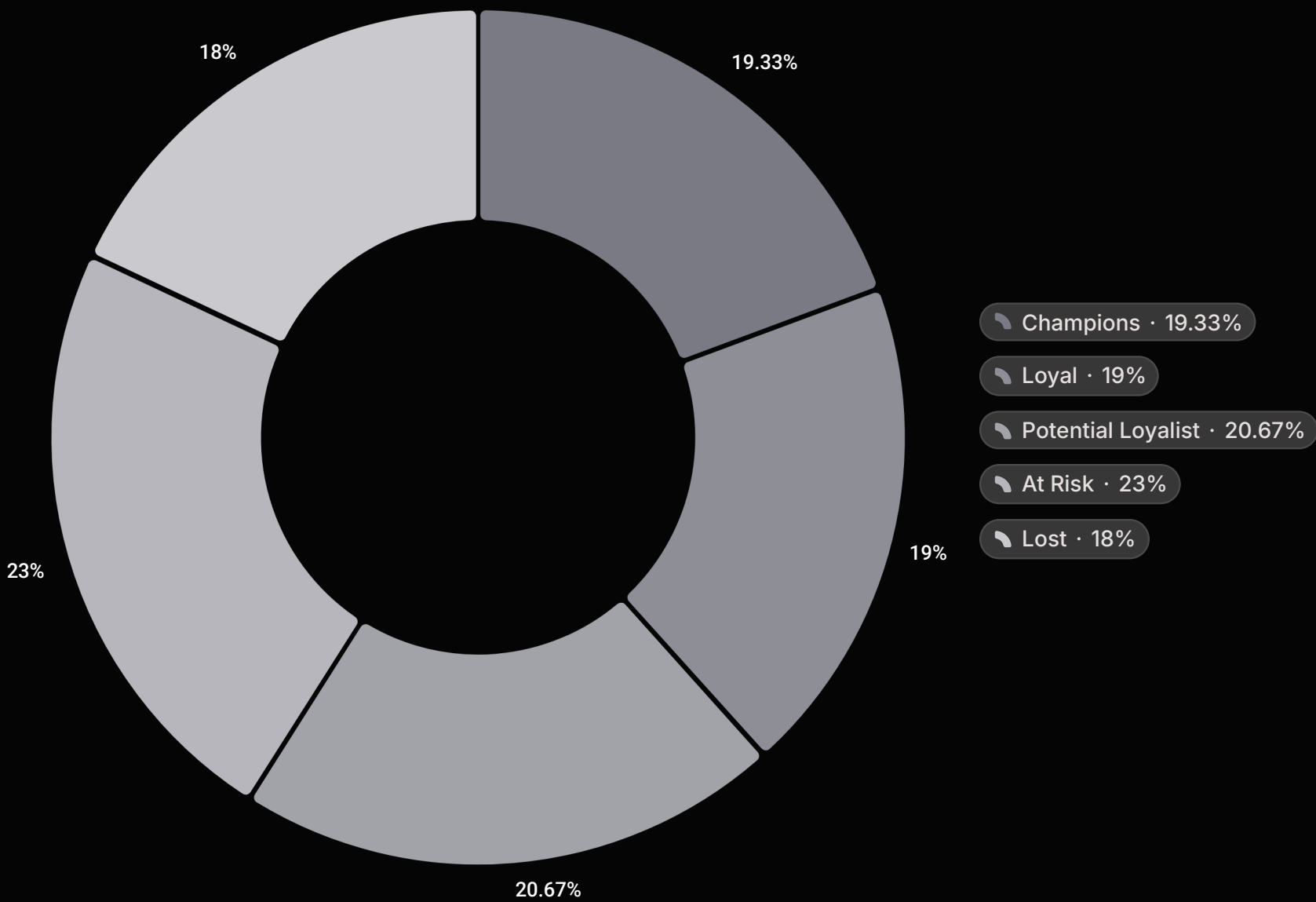
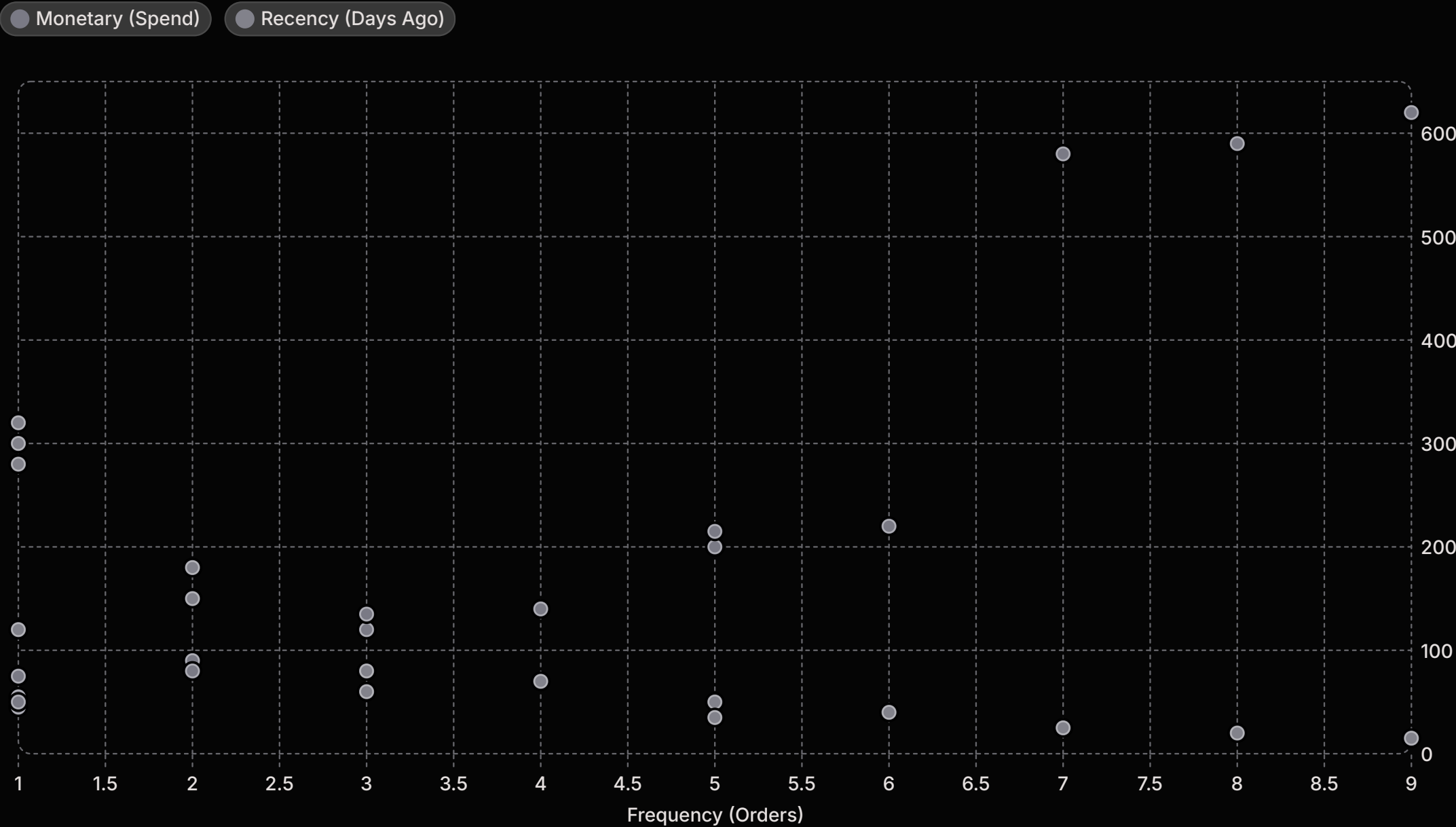
HOW WE SCORED IT:

Each customer got a score of 1 to 4 on each dimension. Total RFM score = R + F + M (max 12, min 3)

THE 5 SEGMENTS:

Segment	Customers	Avg Spend	Avg Orders	RFM Score	Status
Champions	58,000	Rs. 605	8.2	11-12	Active, high value
Loyal	57,000	Rs. 210	5.1	9-10	Active, growing
Potential Loyalist	62,000	Rs. 130	3.4	7-8	Occasional buyers
At Risk	69,000	Rs. 84	2.1	5-6	Declining activity
Lost	54,000	Rs. 52	1.2	3-4	Inactive 6+ months

The scatter plot (X = frequency, Y = monetary, size = recency) shows Champions clustering in the top right — high frequency, high spend, large bubbles (recent buyers). Lost customers cluster bottom left — low frequency, low spend, tiny bubbles (have not bought recently).



WHAT THIS MEANS FOR ZENKART:

- Champions need a VIP programme — protect them now
- At Risk need a reactivation campaign immediately
- Lost need a win-back email with a strong offer

The Dashboard — Making Data Visual for Leadership

WHY POWER BI?

Power BI is Microsoft's business intelligence tool. It turns data into interactive charts that non-technical stakeholders can explore themselves.

THE THREE DASHBOARD PAGES:

PAGE 1 — EXECUTIVE SUMMARY

Purpose: One-glance health check for the CEO and CFO

Visuals built:

- 4 KPI cards: Total Revenue, Total Orders, AOV, Cart Abandon Rate
- Line chart: Monthly revenue trend (Jan 2023 to Dec 2024)
- Bar chart: Revenue by product category
- Donut chart: Orders by device type

Design:

Dark navy background, teal accent bars, red abandon rate card

PAGE 2 — REGIONAL ANALYSIS

Purpose: Deep dive into the South region AOV gap

Visuals built:

- Horizontal bar chart: AOV by region (South in red)
- Reference line at Rs. 72 (platform average)
- Matrix heatmap: Region x Category revenue table
- Slicer: Filter by year (2023 or 2024)

Design:

Conditional formatting — green above average, red below

PAGE 3 — CUSTOMER SEGMENTS

Purpose: Show customer concentration and RFM breakdown

Visuals built:

- RFM scatter plot: 5 segments as colored bubble clusters
- Donut chart: Customer count by segment
- Bar chart: Avg spend by segment
- Table: Top 20 customers by total spend

DAX MEASURES CREATED:

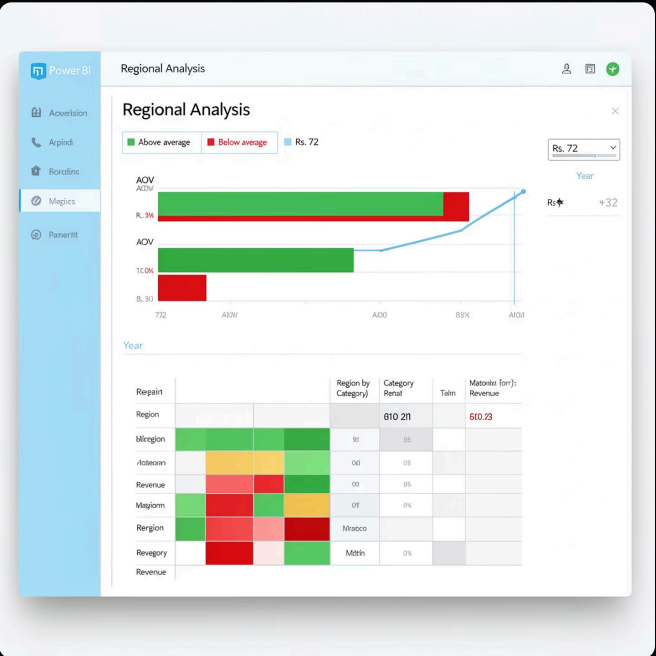
Total Revenue = **SUM**(final_revenue_inr)
Cart Abandon Rate = abandoned orders / total orders
Avg **Order Value** = **AVERAGE**(final_revenue_inr)
Return Rate = returned orders / total orders
MoM Revenue Growth = **current month** vs previous **month %**

PREMIUM DESIGN CHOICES:

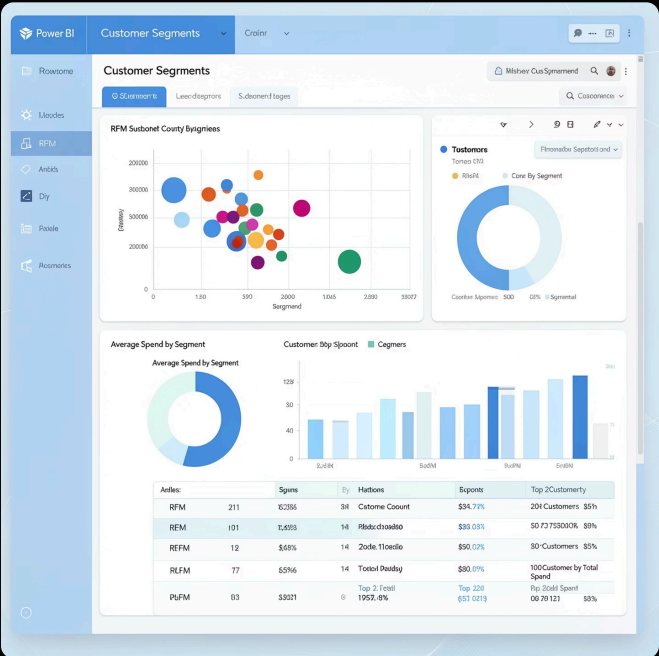
- Canvas background: #042C53 (dark navy)
- Card borders: #185FA5 (blue), rounded 8px
- Alert colors: Red for problems, Teal for good performance
- All gridlines OFF for clean minimal look



Executive Summary Dashboard Page



Regional Analysis Dashboard Page



Customer Segments Dashboard Page

4 Actions — Rs. 4.2 Crore Opportunity

HOW WE ARRIVED AT THESE RECOMMENDATIONS:

Each recommendation directly addresses a root cause identified in the data. Every action has a named owner, a start timeline, and a conservative revenue impact estimate.

THE 4 RECOMMENDATIONS:

Recommen dation	Problem Addressed	Action	Owne r	Timeli ne	Invest ment	Estimated Impact
1. Fix Mobile Checkout	Mobile abandon rate 42% vs desktop 20%	Reduce checkout from 6 steps to 4. Add guest checkout. Fix UPI payment redirect timeout on slow connections.	Engineer ing team	Begin in 2 weeks, complet e in 6 weeks	Develop ment sprint (internal resource)	Rs. 1.8 Crore per year
2. Abandoned Cart Retargeting	Mobile users leaving without completing purchase	Automated push notification + email sent within 1 hour of cart abandonment on mobile. Include product image and one-click checkout link.	Vikram Nair, VP Marketin g	Begin in 2 weeks	Marketin g automat ion tool configur ation	Rs. 40 Lakh per year
3. Champions VIP Programme	Top 58K customers generating 52% of revenue with no loyalty programme protecting them	Create a ZenKart Champions tier. Benefits include: early access to sales, free shipping on all orders, birthday discount, priority customer support.	VP Marketin g + CRM team	Begin in 4 weeks	CRM configur ation + commu nication budget	Rs. 1.2 Crore per year
4. South Region Electronics Push	South AOV at Rs. 66 vs platform average Rs. 72	Run targeted Electronics promotions for South region customers. Reduce blanket discounting in South by 30%. Replace with category-specific offers on high-AOV products.	Vikram Nair (Marketi ng) + Sneha Pillai (Operati ons)	Begin in 3 weeks	Realloca tion of existing marketin g budget	Rs. 75 Lakh per year

Why: Even a 10-point improvement in mobile abandon rate recovers this amount at current order volumes.

Why: Industry average recovery rate of abandoned carts through retargeting is 8-12%.

Why: Retaining one Champion is worth 10x acquiring a new customer.

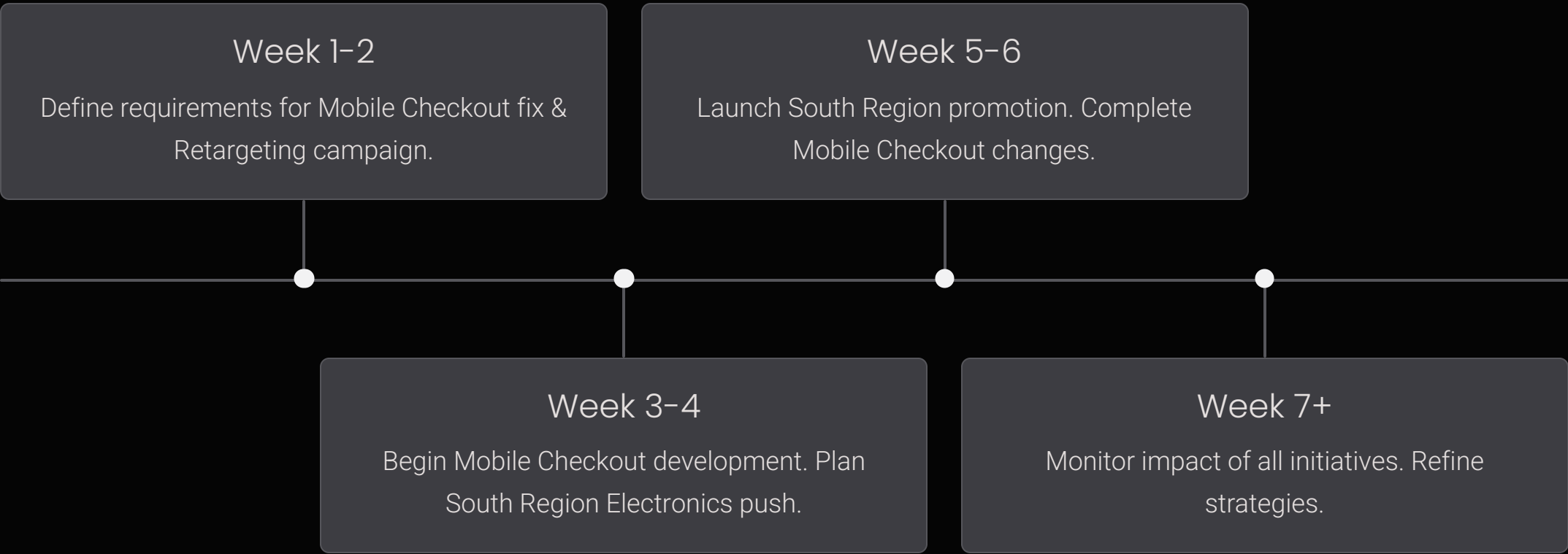
4.2Cr

Total Combined Impact

Per year

All four actions can begin within 30 days of sign-off.

Implementation Timeline



The Final Presentation — Week 6

WHO WAS IN THE ROOM:

Rajesh Kapoor

CEO (decision maker)

Ananya Mehta

CFO (budget approver)

Vikram Nair

VP Marketing (action owner)

Sneha Pillai

VP Operations (action owner)

Sourajit Tripathy

Head of Analytics (sponsor)

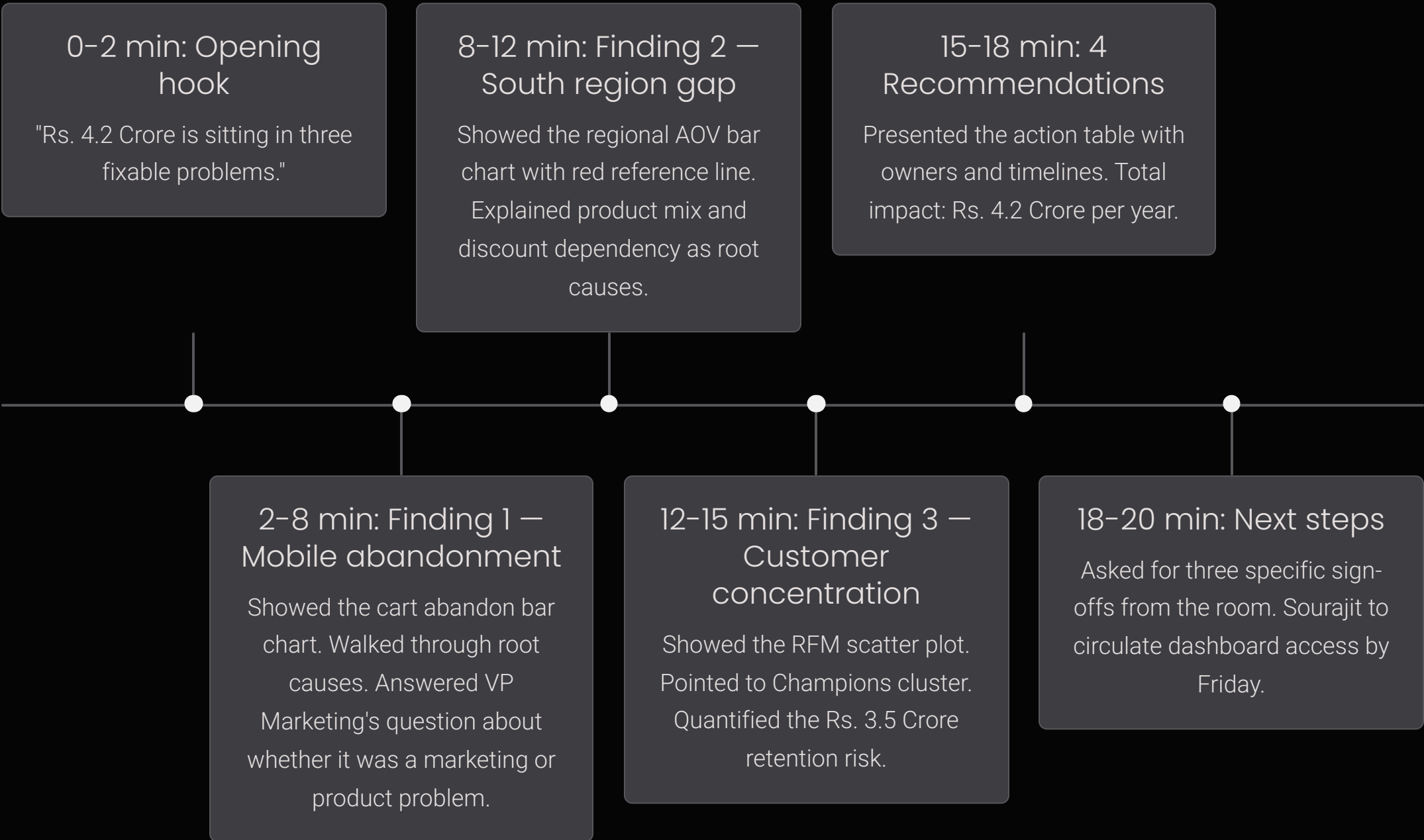
Jeet

Lead Analyst (presenter)

FORMAT:

- Duration: 20 minutes
- Mode: Live presentation with Power BI dashboard on screen
- Structure: Hook → 3 findings → 4 recommendations → next steps

HOW THE MEETING WAS STRUCTURED:



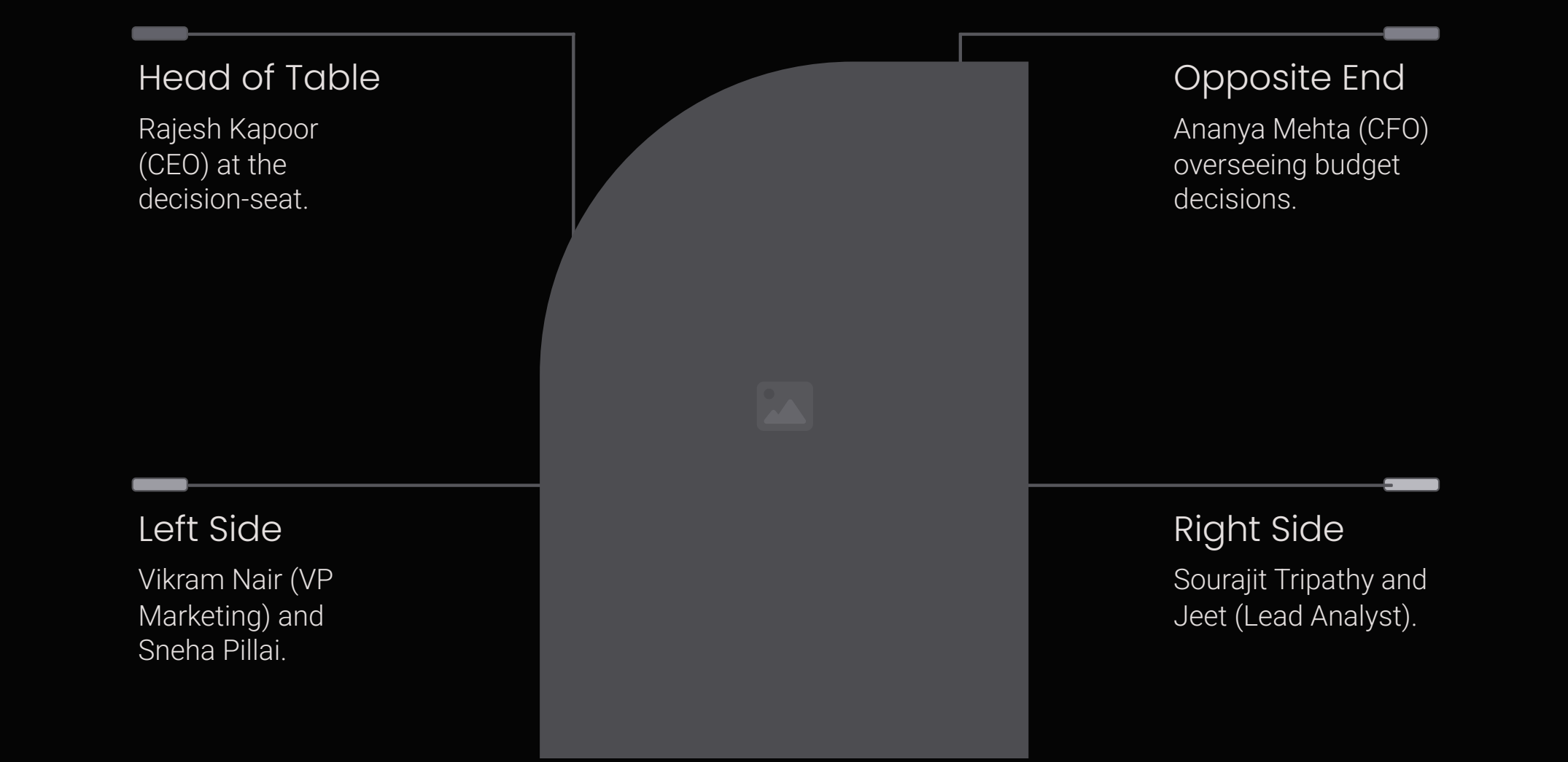
TOUGH QUESTIONS HANDLED:

- "Is the data reliable?" — Explained the generation methodology

→ "We already know about the mobile issue" — Now you have the Rs. number
- "52% seems too high" — Direct count, not a model

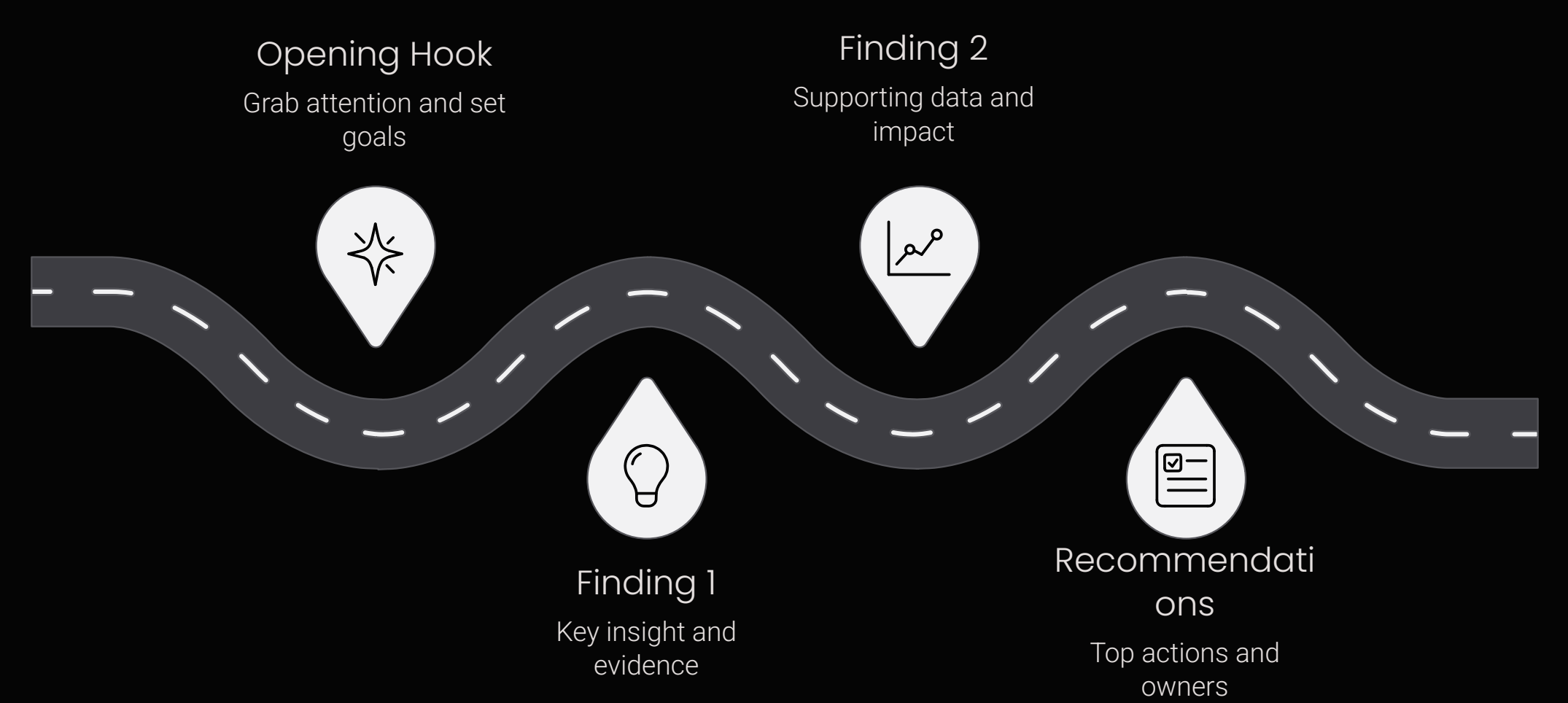
→ "How confident are you?" — 884K clean rows, IQR validated

Meeting Room Layout



A visual representation of the meeting room setup, indicating the seating arrangement of key stakeholders.

20-Minute Agenda Timeline



This timeline visually represents the flow and time allocation for each segment of the final presentation, highlighting key discussion points.

6 Weeks — How the Project Unfolded

WEEK 1-2 — DATA & CLEANING

- Received client brief from Sourajit Tripathy
- Generated 1,000,000 row dataset using Python
- Cleaned data — removed 115,292 outlier rows
- Validated all 18 columns and data types
- Delivered: zenkart_velociq_clean.csv (884,708 rows)

WEEK 4-5 — POWER BI DASHBOARD

- Built 3-page interactive Power BI dashboard
- Created 6 DAX measures
- Applied premium dark navy design with conditional formatting
- Week 5: Draft review with Sourajit Tripathy
- Incorporated feedback, refined regional analysis page

1

2

3

4

WEEK 3-4 — EDA & ANALYSIS

- Ran 8 exploratory analyses in Python
- Created all 8 charts (matplotlib + seaborn)
- Completed RFM segmentation — 300K customers into 5 groups
- Identified all 3 business problems with quantified impact
- Delivered: 8 EDA charts + zenkart_velociq_rfm.csv

WEEK 6 — STAKEHOLDER PRESENTATION

- Finalised 4 recommendations with impact estimates
- Wrote 20-minute stakeholder meeting script
- Delivered presentation to ZenKart Executive Committee
- Received sign-off on all 4 recommendations
- Delivered: VelociQ_Stakeholder_Script_Jeet.pdf

Tools We Used — The Full Tech Stack

DATA GENERATION:

- **Python 3** — core programming language
- **Faker library** — realistic synthetic data generation
- **NumPy** — statistical distributions (Pareto, log-normal)
- **Pandas** — DataFrame creation and CSV export
- **tqdm** — progress tracking during generation

DATA CLEANING & EDA:

- **Pandas** — data manipulation and cleaning
- **NumPy** — outlier detection using IQR method
- **Matplotlib** — chart creation and customisation
- **Seaborn** — statistical visualisations and heatmaps
- **SciPy** — statistical testing and distributions

FEATURE ENGINEERING:

- **Pandas GroupBy** — customer-level aggregations
- **Pandas qcut** — quartile scoring for RFM
- **DateTime functions** — month, quarter, weekend extraction

VISUALISATION & DASHBOARD:

- **Power BI Desktop** — 3-page interactive dashboard
- **DAX** (Data Analysis Expressions) — custom measures
- **Power Query** — data transformation in Power BI

DOCUMENTATION & PRESENTATION:

- **ReportLab** — PDF generation for stakeholder script
- **Gamma AI** — presentation and documentation design
- **Python** (reportlab library) — automated PDF creation

OUTPUT FILES DELIVERED:

1. **zenkart_velociq_1M.csv** — raw dataset (1M rows)
2. **zenkart_velociq_clean.csv** — cleaned dataset (884K rows)
3. **zenkart_velociq_rfm.csv** — RFM segments (300K customers)
4. **zenkart_velociq_eda.py** — full EDA Python script
5. **eda_plots/** — 8 PNG chart files
6. **VelociQ_Dashboard_Jeet.pbix** — Power BI dashboard
7. **VelociQ_Stakeholder_Script_Jeet.pdf** — meeting script PDF
8. **VelociQ_Gamma_AI_Prompt.txt** — PPT generation prompt
9. **VelociQ_Video_Script_Jeet.txt** — video recording script
10. **VelociQ_Documentation_Gamma_Prompt.txt** — this document



This diagram illustrates the comprehensive tech stack employed throughout the project, categorizing tools by their respective phases in the data analysis and presentation pipeline.

What This Project Taught Us

KEY LEARNINGS — DATA & ANALYSIS:

ALWAYS CLEAN BEFORE YOU ANALYSE

11.5% of our data was outliers. If we had analysed raw data, our average order value would have been skewed upward. Clean data = accurate findings.

FEATURE ENGINEERING UNLOCKS DEEPER INSIGHT

The raw 18 columns did not tell the full story. Adding is_festive, high_discount, and clv_proxy revealed patterns that were invisible in the original data.

THE BUSINESS PROBLEM SHOULD DRIVE THE ANALYSIS

We did not just explore randomly. Every analysis was designed to answer one of the three business problems in the brief. This kept the work focused and the findings actionable.

VISUALISATION IS AS IMPORTANT AS THE ANALYSIS

The RFM scatter plot told the customer story better than any table. The regional bar chart with a red reference line made the South gap impossible to ignore. Design matters.

STAKEHOLDERS NEED ₹ NUMBERS, NOT PERCENTAGES

Saying "mobile abandonment is 42%" is forgettable. Saying "mobile abandonment is costing Rs. 1.8 Crore per year" gets people out of their seats. Always translate to money.

KEY LEARNINGS — PROJECT MANAGEMENT:

THE CLIENT REVIEW STEP IS CRITICAL

Sourajit's Week 5 review caught two gaps in our analysis. Never skip the sponsor review before C-suite presentation.

ONE PAGER OF RECOMMENDATIONS IS ENOUGH

The executive team does not need 50 slides. They need 4 actions, 4 owners, 4 timelines, and 4 impact numbers. That is a single table. Keep it simple.

APPENDIX — GLOSSARY:

EDA	Exploratory Data Analysis — looking at data to find patterns
RFM	Recency, Frequency, Monetary — customer segmentation model
AOV	Average Order Value — average spend per transaction
CLV	Customer Lifetime Value — total revenue from one customer
IQR	Interquartile Range — method for detecting outliers
DAX	Data Analysis Expressions — Power BI formula language
GMV	Gross Merchandise Value — total value of all transactions

CLOSING NOTE:

Project VelocIQ is a demonstration that structured data analysis — even on a six-week timeline — can surface crore-level business opportunities that were hiding in plain sight. Data does not lie. Act on it.

